Question	Answer	Marks	Partial Marks
9(b)	(3, 27)	**3**	**B2** for $x = 3$ OR **M1** for *their* $12x^2 - 4x^3 = 0$ or better or states $\frac{dy}{dx} = 0$ **M1dep** for substituting *their x* into $y = 4x^3 - x^4$ shown
9(c)	–64	**3**	**M1** for $4x^3 - x^4 = 0$ **B1** for $x = 4$
10(a)	[*DEF*], *BCD* *ADF*, *ADB*	**2**	**B1** for each pair
10(b)	*OQ* *OQT* Tangent perpendicular to radius RHS equal	**5**	**B1** for each
11(a)	4	**1**	
11(b)	$7 - 3x$ final answer	**2**	**M1** for $1 - 3(x - 2)$
11(c)	$\frac{1-x}{3}$ oe final answer	**2**	**M1** for $x = 1 - 3y$ or $y - 1 = -3x$ or $1 - y = 3x$ or $\frac{y}{3} = \frac{1}{3} - x$
11(d)	$a = 2$, $b = 5$, $c = -1$	**5**	**B4** for two correct values <u>only</u> after correct substitution seen i.e. $(1 - 3x - 1)^2 - (x - 1)^2(1 - 3x)$ or for correct unsimplified expansion or a correct simplified expansion. OR **M1** for $(1 - 3x - 1)^2 - (x - 1)^2(1 - 3x)$ **B2** for correct expansion of $[-](x - 1)^2(1 - 3x)$ $[-](x^2 - x - x + 1 - 3x^3 + 3x^2 + 3x^2 - 3x)$ or better or **B1** for expansion of one pair of brackets $\left[(x-1)^2 = \right]x^2 - x - x + 1$ or better or $[(x - 1)(1 - 3x) =] - 3x^2 + x + 3x - 1$

Question	Answer	Marks	Partial Marks
11(e)	$\frac{3-x+3x^2}{x}$ final answer	**3**	**B1** for $3-x(1-3x)$ or better **B1** for common denominator x isw
11(f)	–7	**1**	
12(a)(i)	$\begin{pmatrix}2\\5\end{pmatrix}$	**1**	
12(a)(ii)	$\begin{pmatrix}-6\\4\end{pmatrix}$	**1**	
12(b)	$[y=]-\frac{2}{3}x+\frac{19}{3}$ oe	**3**	**M1** for gradient = $\frac{1-5}{8-2}$ oe **M1** for substituting (8, 1) or (2, 5) into *y = their mx + c*
12(c)	$[y=]\frac{3}{2}x-\frac{9}{2}$ oe	**4**	**B1** for (5, 3) oe **M1** for gradient = $-\frac{1}{\textit{their}\text{ gradient of } AB}$ **M1** substituting *their* midpoint into *y = their mx + c*
12(d)	$\frac{65}{6}$ oe	**2**	**M1** for *their* $\frac{19}{3}$ – *their* $-\frac{9}{2}$ oe